

SQL Functions — Revision Notes

Gate Smashers Lesson • Single-Row vs Multi-Row Functions

EXAM NOTES

1. WHAT IS A SQL FUNCTION?

A SQL function is a built-in operation that processes input data and returns a computed result. SQL functions are broadly classified into two categories:

1. Single-Row Functions

Operate on **one row at a time** and return **one output per row**.

2. Multi-Row Functions

Operate on **multiple rows** (group of rows) and return **one aggregated output**.

2. SINGLE-ROW FUNCTIONS

Definition: Applied to each row individually in a table query result set.

 **Memory Rule:** Single Row → Single Input → Single Output (1 Input Row = 1 Output Row)

Categories of Single-Row Functions

N C D G C ⇒ Number • Character • Date • General • Conversion

① Number Functions

Function	Purpose	Example & Behavior
ROUND	Rounds a number to specified decimal places	ROUND(50.926, 2) ⇒ 50.93 (3rd digit ≥ 6)
TRUNC	Truncates/removes decimal portion without rounding	TRUNC(50.926, 2) ⇒ 50.92
MOD	Returns remainder of division	MOD(10, 3) ⇒ 1 (10 ÷ 3 remainder)

② Character Functions

Operate on text/string values:

Function	Description & Example	Result
LOWER	Converts string to lowercase: LOWER('HUSEN')	husen
UPPER	Converts string to uppercase: UPPER('husen')	HUSEN
INITCAP	Capitalizes first letter of each word: INITCAP('husen basha')	Husen Basha
CONCAT	Combines two strings: CONCAT('Husen', 'Basha')	HusenBasha
LENGTH	Finds total character count: LENGTH('HUSEN')	5
SUBSTR	Extracts substring from a larger string	SUBSTR('HUSEN', 1, 3) ⇒ HUS
INSTR	Returns character position inside string (IN String)	Finds index of character
LPAD / RPAD	Pads string with characters on Left / Right side	Formatted string
TRIM	Removes leading/trailing spaces or characters	Cleaned string
REPLACE	Replaces target portion of string with new value	Replaced string

③ Date Functions

- MONTHS_BETWEEN : Calculates months between 2 dates
- ADD_MONTHS : Adds specified months to a date
- LAST_DAY : Returns last day of current date's month
- ROUND / TRUNC : Rounds or truncates dates to unit (e.g.

④ General Functions ⚙️

Used primarily for **NULL value handling** and conditional evaluation logic.

Function	Purpose & Behavior
<code>NVL(exp1, exp2)</code>	Replaces <code>NULL</code> value in <code>exp1</code> with <code>exp2</code> .
<code>NVL2(exp1, exp2, exp3)</code>	Returns <code>exp2</code> if <code>exp1</code> is NOT NULL ; returns <code>exp3</code> if <code>exp1</code> is NULL .
<code>COALESCE(...)</code>	Scans list of parameters and returns the first non-NULL value.
<code>CASE</code>	Provides conditional IF-THEN-ELSE logic within SQL statements.
<code>DECODE</code>	Oracle-specific function providing value-matching logic similar to <code>CASE</code> / switch.

⑤ Data Type Conversion Functions ➡️

Converts data from one data type to another (similar to type casting in programming):

<code>TO_CHAR</code> Number/Date → Character	<code>TO_NUMBER</code> Character → Number	<code>TO_DATE</code> Character → Date
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3. MULTI-ROW FUNCTIONS (AGGREGATE FUNCTIONS)

Definition: Operates across a set of multiple rows to compute a single aggregated value.

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-- Example Query
SELECT SUM(salary)
FROM employee;

-- If salary values are: 20000, 30000, 40000
-- Output: 90000 (1 single row output)
```

S M M C A

- `SUM()` – Total sum of values
- `MAX()` – Highest value in column
- `MIN()` – Lowest value in column
- `COUNT()` – Total count of rows / non-null entries
- `AVG()` – Calculated arithmetic mean

4. MAJOR DIFFERENCE: SINGLE-ROW VS MULTI-ROW

Feature	Single-Row Function	Multi-Row (Aggregate) Function
Input	Processes 1 row at a time	Processes multiple rows together
Output	1 result for every row	1 result for the <i>entire</i> set of rows
Also Known As	Scalar Functions	Aggregate Functions / Group Functions
Examples	<code>UPPER()</code> , <code>ROUND()</code> , <code>LENGTH()</code> , <code>NVL()</code>	<code>SUM()</code> , <code>MAX()</code> , <code>MIN()</code> , <code>AVG()</code> , <code>COUNT()</code>

⚠️ **Exam Trap**

Do not confuse individual row modifiers like `ROUND()` with aggregate summary functions like `SUM()`. `ROUND()` modifies each row value independently, whereas `SUM()` accumulates values across rows.

🧠 QUICK REVISION SUMMARY

★ **Golden Rule:**

- Single-Row:** 1 Input Row ⇒ 1 Output Result
- Multi-Row:** Many Input Rows ⇒ 1 Aggregate Output Result

One-Line Summary: Single-row functions process each row individually, whereas multi-row/aggregate functions process multiple rows to return one combined summary result.

